

BISHA, Saudi Arabia

WMO: 410840

Lat: 19.984N

Lon: 42.621E

Elev: 1185

StdP: 87.88

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	8.8	-13.0	1.4	19.1	-10.1	1.8	21.6	8.2	22.9	7.2	21.9	0.5	180	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	40.2	18.0	40.0	17.8	39.2	17.5	21.5	31.2	20.8	31.2	20.1	31.5	3.7	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	15.9	24.6	17.9	14.9	24.6	17.0	14.0	24.3	68.7	31.1	65.7	31.0	63.3	31.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.9	6.0	DB	3.7	41.7	1.9	0.6	2.3	42.1	1.2	42.4	0.1	42.7	-1.3	43.1	(4)
(5)			WB	-1.6	22.8	2.0	1.0	-3.0	23.5	-4.2	24.1	-5.3	24.6	-6.8	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	17.8	20.4	23.6	26.4	29.6	31.8	32.8	32.6	29.8	25.0	21.6	18.2	(6)
(7)		DBStd	5.70	2.63	2.64	2.58	2.40	2.02	1.52	1.53	1.35	1.89	1.71	1.93	2.32	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	87	41	10	1	1	0	0	0	0	0	0	2	32	(9)
(10)		CDD10.0	5778	242	292	422	491	607	654	708	701	594	465	349	253	(10)
(11)		CDD18.3	2823	24	69	165	242	348	404	450	443	344	207	101	26	(11)
(12)		CDH23.3	41352	435	886	1876	2843	4938	6556	7311	7112	5153	2673	1144	426	(12)
(13)	CDH26.7	24749	97	308	832	1494	2986	4346	4913	4746	3229	1351	376	71	(13)	
(14)	Wind	WSAvg	2.2	1.8	2.2	2.4	2.4	2.3	2.2	2.5	2.4	2.3	2.6	2.0	1.7	(14)
(15)	Precipitation	PrecAvg	89	10	2	11	32	13	3	2	3	1	2	7	3	(15)
(16)		PrecMax	276	94	11	58	151	94	33	34	13	18	24	38	23	(16)
(17)		PrecMin	31	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	59	19	3	14	32	19	8	6	3	3	5	11	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	33.2	35.5	37.1	39.2	41.0	41.2	41.1	40.0	35.9	32.2	30.1	(19)
MCWB			15.2	16.1	17.3	17.2	17.6	18.1	18.6	18.7	18.2	16.3	16.2	15.4	(20)	
2%		DB	29.1	31.9	33.9	36.0	38.2	40.1	40.2	40.2	38.9	34.7	31.2	28.5	(21)	
		MCWB	15.3	16.0	16.5	17.1	17.6	17.6	17.9	18.3	17.5	15.8	15.7	15.4	(22)	
5%		DB	27.5	30.2	32.8	34.9	37.8	39.2	39.9	39.8	38.0	33.7	30.2	27.1	(23)	
		MCWB	15.1	15.8	16.9	17.1	17.6	17.3	17.7	18.1	17.0	15.3	15.6	14.9	(24)	
10%		DB	25.9	28.8	31.1	33.8	36.7	38.8	39.0	39.0	37.1	32.8	29.0	26.0	(25)	
		MCWB	14.6	15.3	16.5	17.1	17.3	17.1	17.5	17.7	16.5	15.1	15.3	14.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	19.2	20.3	21.1	21.6	21.1	22.2	22.7	20.6	20.3	19.5	19.1	(27)
MCDWB			26.8	27.5	30.6	29.4	30.3	32.3	33.0	31.6	33.0	26.0	26.6	26.8	(28)	
2%		WB	17.3	18.1	19.5	20.3	20.7	19.9	21.1	21.7	19.3	18.8	18.5	17.9	(29)	
		MCDWB	25.6	27.8	29.3	29.0	29.8	34.1	32.2	31.3	34.1	26.7	26.2	25.1	(30)	
5%		WB	16.4	17.4	18.8	19.6	19.9	19.2	20.0	20.8	18.5	17.8	17.7	16.9	(31)	
		MCDWB	24.7	27.1	28.2	28.4	29.8	34.5	33.9	32.1	34.3	27.5	25.9	24.4	(32)	
10%		WB	15.6	16.5	18.0	19.1	19.3	18.5	19.2	19.8	17.7	16.6	17.0	15.7	(33)	
		MCDWB	24.0	26.1	27.6	28.0	30.0	34.7	34.7	33.7	34.5	29.2	25.3	23.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.9	16.0	15.1	14.0	14.0	14.4	13.0	13.6	15.1	15.5	15.0	15.9	(35)
MCDBR			18.0	17.7	16.4	15.2	14.3	14.6	13.3	13.9	15.1	15.9	16.2	17.1	(36)	
MCWBR			8.8	8.2	6.9	5.5	4.9	5.3	5.2	5.4	6.1	6.9	7.6	8.6	(37)	
5% WB		MCDBR	15.6	15.6	14.4	13.6	13.7	13.9	12.6	12.9	13.9	13.6	13.7	14.5	(38)	
		MCWBR	8.1	7.9	6.3	5.0	4.4	5.5	5.5	5.5	5.9	5.8	6.4	7.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.414	0.464	0.512	0.557	0.585	0.623	0.629	0.531	0.421	0.405	0.376	(40)	
taud		2.221	2.084	1.949	1.856	1.732	1.652	1.617	1.602	1.766	2.045	2.118	2.212	(41)		
Ebn at Noon		891	877	847	808	765	737	710	709	777	858	853	878	(42)		
Edh at Noon		131	159	190	210	236	253	263	268	224	163	144	129	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.01	5.79	6.51	6.81	7.29	7.56	7.09	6.84	6.74	6.32	5.29	4.85	(44)	
(45)		RadStd	0.23	0.23	0.27	0.35	0.35	0.20	0.25	0.17	0.17	0.18	0.20	0.23	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+1.01	+1.17	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	+0.48	N/A	N/A	+0.57	+0.32	+0.21	N/A	-73	+200	+126

Nomenclature: See separate page